

→

$$\frac{1}{\lambda} = \frac{1}{\lambda_0} - \frac{v}{mc^2}$$

$$\Delta \lambda \propto \phi$$

→ UNIT → 2

→ wave nature of particles

• de Broglie wavelength

→ suggested in 1925 about wave nature

→ verified by Davisson & Germer in 1927

$$\lambda = \frac{h}{mv}$$

$$mv \propto \frac{1}{\lambda}$$

"Momentum of particle is inversely proportional to wavelength of wave associated with it"

→ derivation

if it's a photon of wave freq ν , then

$$E = h\nu \quad - (1)$$

if it's a mass 'm' particle then

$$E = mc^2 \quad - (2)$$

From (1) & (2)

$$h\nu = mc^2 \quad - (3)$$

also $mc = p \rightarrow \text{momentum}$
 $p = mc \quad - (4)$

$$\frac{h\nu}{p} = \frac{mc^2}{mc}$$

$$\frac{h\nu}{c} = p$$

$$\frac{h}{p} = \frac{c}{\nu}$$

$$\Rightarrow \frac{h}{p} = \lambda$$

$$\Rightarrow \lambda = \frac{h}{mv}$$

de Broglie Wavelength

→ derivation using ψ

consider a particle as standing wave system
 at any instant t_0 at pt. (x_0, y_0, z_0) be represented by the function

$$\psi = \psi_0 \sin 2\pi \nu_0 t_0$$

$\nu_0 \rightarrow$ frequency ψ_0 is amplitude at (x_0, y_0, z_0)

if it's given a velocity in x -dirⁿ then by inverse Lorentz transformation,

$$t_0 = t + \frac{Vx}{c^2}$$

$$\lambda_0 = \lambda \sqrt{1 - \frac{V^2}{c^2}}$$

$$\lambda = \frac{\lambda_0}{\sqrt{1 - \frac{V^2}{c^2}}}$$

& variation of ψ is given by

$$\psi = \psi_0 \sin \left[2\pi \nu_0 \left[\frac{t + \frac{v}{c^2} x}{\sqrt{1 - \frac{v^2}{c^2}}} \right] \right] \quad - (1)$$

Also, standard eq of wave motion is

$$\psi = \psi_0 \sin \left[\frac{2\pi}{T} \left(t + \frac{x}{u} \right) \right] \quad - (2)$$

Amplitude

Periodic time

Phase velocity along x-axis

Comparing (1) & (2)

$$u = \frac{c^2}{v} ; \quad \frac{1}{T} = \nu_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$\text{Also, } E = mc^2 = h\nu_0$$

$$\Rightarrow \nu_0 = \frac{mc^2}{h}$$

$$\Rightarrow v = \frac{mc^2}{h} \sqrt{1 - \frac{v^2}{c^2}}$$

$$\text{Also, } m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow v = \frac{mc^2}{h}$$

also, $\lambda = \frac{\text{velocity}}{\text{frequency}} = \frac{u}{\nu}$

$$\Rightarrow \lambda = \frac{c^2}{v} \frac{h}{mc^2}$$

$$\Rightarrow \lambda = \frac{h}{mv}$$

$$\Rightarrow \lambda = \frac{h}{p}$$

also, $p = \sqrt{2mKE}$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mKE}} \quad \left[\text{if } KE \rightarrow \text{kinetic energy} \right]$$

→ if charged to q , accelerated via potential diff V ,

$$KE = qV$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mqV}}$$

for an e^-

$$\lambda = \frac{12.27}{\sqrt{V}} \quad \rightarrow \text{potential diff}$$

UNCERTAINTY PRINCIPLE

It is impossible to measure simultaneously the value of 2 conjugate canonical variables with unlimited accuracy.

2 variables can be :-

1. position, momentum
2. energy, time
3. angular momentum, angle
- 4.

$$\Rightarrow \Delta x \Delta p_x \geq \frac{\hbar}{2} \quad \left(\hbar = \frac{h}{2\pi} \right)$$

$$\Rightarrow \Delta E \Delta t \geq \frac{\hbar}{2}$$

$$\Rightarrow \Delta J \Delta \theta \geq \frac{\hbar}{2}$$

where Δn represents uncertainty in the measurement of variable 'n'

$$\Delta x \Delta p_x \geq \frac{\hbar}{2} \quad \Delta y \Delta p_y \geq \frac{\hbar}{2} \quad \Delta z \Delta p_z \geq \frac{\hbar}{2}$$

$$\text{but } \Delta x \Delta p_y = 0 \text{ or } \Delta y \Delta p_z = 0 \\ \text{or } \Delta z \Delta p_x = 0$$

if both are in same dirⁿ, tab hi hoga $\geq \frac{\hbar}{2}$ warna 0 hoga.

$$\hat{x} = x$$

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$$

$$\hat{E}_x = -i\hbar \frac{\partial}{\partial t}$$

→ APPLICATIONS OF UNCERTAINTY PRINCIPLE

① NON EXISTENCE OF ELECTRONS IN NUCLEUS

Radius of nucleus of any atom is of the order of 10^{-14} m. so if e^- is inside the nucleus, $\Delta x \geq 10^{-14}$ m.

$$\text{Now, } \Delta x \Delta p \geq \frac{h}{2}$$

$$\Delta p = \frac{h}{2\Delta x}$$

$$\text{Also } \Delta x = 2 \times 10^{-14} \text{ m} \quad (2R)$$

$$\Delta p = \frac{6.63 \times 10^{-34}}{2 \times 2 \times 10^{-14}}$$

$$\Delta p = 5.275 \times 10^{-21} \text{ kg m/s}$$

$$\Rightarrow p \approx 5.275 \times 10^{-21} \text{ kg m/s}$$

$$\text{Also, } KE = \frac{p^2}{2m} = \frac{5.275 \times 5.275 \times 10^{-42}}{2 \times 9.1 \times 10^{-31}} \text{ J}$$

$$KE = 1.528 \times 10^{-11} \text{ J}$$

$$KE = \frac{1.528 \times 10^{-11}}{1.6 \times 10^{-19}} \text{ eV}$$

$$KE = 0.955 \times 10^8 \text{ eV}$$

$$KE = 95.5 \text{ MeV}$$

$$KE \approx 97 \text{ MeV}$$

So, if e^- exists in nucleus the $KE = 97 \text{ MeV}$ but e^- has very small energy of 4 MeV hence, e^- doesn't exist inside nucleus.

② DETERMINATION OF SIZE OF ATOM

Consider H atom with a e^- & proton. Again assuming that e^- is confined to a region $\sim a$ then acc to uncertainty principle

$$p \sim \Delta p \sim \frac{h}{2a}$$

$$\frac{10^{-68}}{10^{-31} \times 10^9 \times 10^{-38} \times \frac{1}{180}} = \frac{1}{180}$$

$$KE = \frac{\Delta p^2}{2m} \approx \frac{h^2}{8ma^2}$$

$$PE \propto -\frac{ke^2}{a}$$

$$TE \approx \frac{h^2}{8ma^2} - \frac{ke^2}{a}$$

Minimise w.r.t a

$$\frac{dTE}{da} = 0$$

$$\Rightarrow -\frac{h^2}{4ma^3} + \frac{ke^2}{a^2} = 0$$

$$\Rightarrow a \approx \frac{h^2}{mke^2} \approx a_0$$

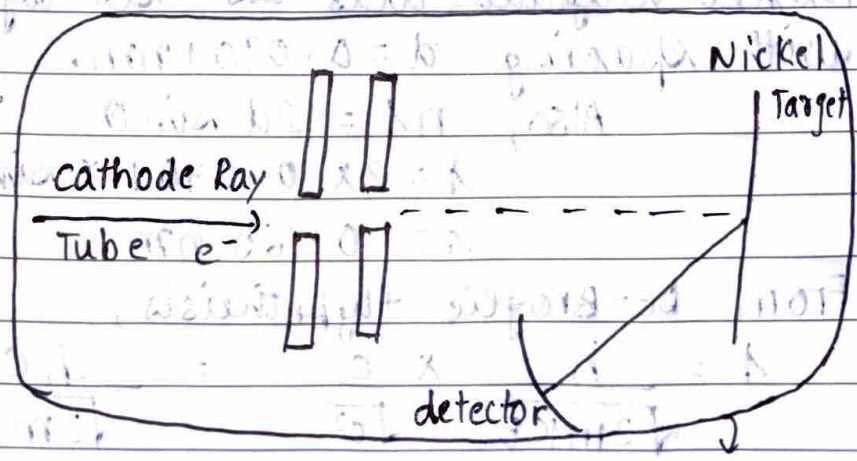
→ DAVVISON AND GERMER EXPERIMENT

1925 → de Broglie → wave nature

1927 → Davison Germer → experimentally proved wave nature of matter

• SETUP

vacuum chamber, cathode ray tube, nickel crystal & detector



vacuum chamber

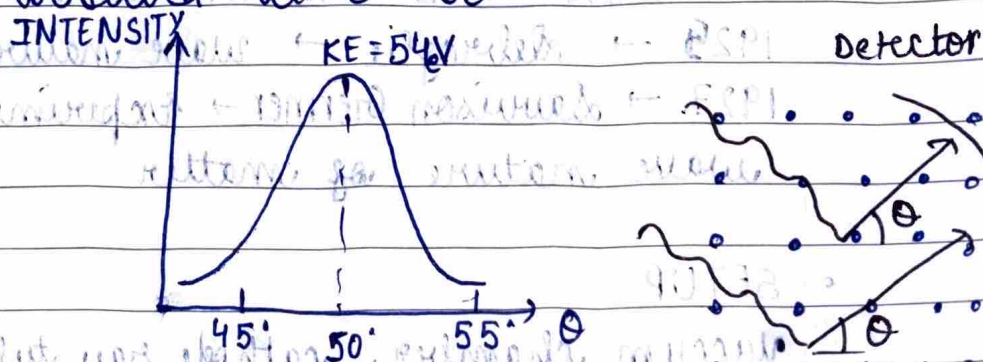
• WORKING

$$1 = \frac{h}{\sqrt{2mKE}}$$

- When e^- beam strikes nickel crystal it's scattered in different directions.
- Intensity is measured as funcⁿ of θ

• OBSERVATION

- At certain voltages & angles, the intensity of scattered e^- showed sharp maxima lly to X ray diffraction pattern.
- For accelerating $KE = 54\text{eV}$ the max intensity was observed at $\theta = 50^\circ$



• VERIFICATION USING BRAGG'S LAW

Nickel crystal acts as set of parallel planes with spacing $d = 0.0909\text{ nm}$.

$$\text{Also, } n\lambda = 2d \sin \theta$$

$$\lambda = 2 \times 0.0909 \times \sin 50$$

$$\lambda = 0.165\text{ nm}$$

From De-Broglie Hypothesis,

$$\lambda = \frac{h}{\sqrt{2mKE}} \times \frac{c}{\sqrt{c^2}} = \frac{hc}{\sqrt{2mc^2 KE}}$$

$\therefore e^-$ diffracted similar as X-rays

$$\Rightarrow \lambda = \frac{1240 \text{ eV nm}}{\sqrt{2 \times 511 \text{ KeV} \times 54 \text{ eV}}}$$

$$\Rightarrow \lambda \approx 0.166 \text{ nm}$$

$$\lambda_{\text{theoretical}} = 0.166 \text{ nm}$$

$$\lambda_{\text{experimental}} = 0.165 \text{ nm}$$

Hence de-Broglie was proved right & wave nature was exp. proven.

→ BOHR'S POSTULATES

$$1. \quad L = mvr = \frac{nh}{2\pi}$$

Angular momentum is an integral multiple of $\frac{h}{2\pi}$

$$2. \quad \frac{mv^2}{r} = \frac{ze^2}{4\pi\epsilon_0 r^2}$$

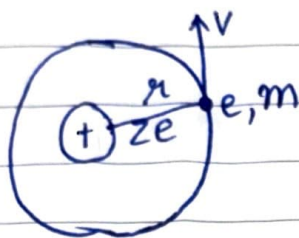
centripetal force of circular motion is equal to coulomb force.

$$3. \quad E_f - E_i = h\nu$$

Difference between 2 energies of 2 states is equal to $h\nu$

4. Stationary orbits doesn't radiates energy.

→ BOHR'S MODEL FOR HYDROGEN ATOM



According to classical mechanics

Consider an electron with charge $-e$ & mass m revolving around a nucleus of atom with atomic no z and charge $+ze$ of radius r .

It exhibits circular motion

Centripetal force = Coulomb's attractive force

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{ze(e)}{r^2}$$

$$\Rightarrow v^2 = \frac{1}{4\pi\epsilon_0} \frac{ze^2}{mr} \quad \text{--- (1)}$$

$$\text{Also, } mvr = \frac{nh}{2\pi}$$

$$\Rightarrow v = \frac{nh}{2\pi mr} \quad \text{--- (2)}$$

$$\Rightarrow v^2 = \frac{n^2 h^2}{4\pi^2 m^2 r^2}$$

Put in (1)

$$\Rightarrow \frac{n^2 h^2}{4\pi^2 m^2 r^2} = \frac{1}{4\pi\epsilon_0} \frac{ze^2}{r}$$

$$\Rightarrow \frac{n^2 h^2}{\pi m r} = \frac{ze^2}{\epsilon_0}$$

$$\Rightarrow r = \frac{n^2 h^2 \epsilon_0}{ze^2 \pi m} \quad \text{--- (3)}$$

$$\Rightarrow r = \frac{6.63 \times 6.8 \times 10^{-68} \times 8.85 \times 10^{-12}}{1 \times 1.6 \times 1.6 \times 10^{-38} \times 3.14 \times 9.1 \times 10^{-31}} \times \frac{n^2}{Z}$$

$$\Rightarrow r = 0.529 \times \frac{n^2}{Z} \times 10^{-10} \text{ m}$$

$$\Rightarrow r = 0.529 \times \frac{n^2}{Z} \text{ \AA}$$

using (3) in (2)

$$v = \frac{2\pi h}{2\pi m \frac{n^2 h^2 \epsilon_0}{ze^2 m}}$$

$$\Rightarrow v = \frac{ze^2}{2nh \epsilon_0} \quad (4)$$

$$\Rightarrow v = \frac{1.6 \times 1.6 \times 10^{-38}}{2 \times 6.63 \times 10^{-34} \times 8.85 \times 10^{-12}} \times \frac{Z}{n}$$

$$\Rightarrow v = 2.18 \times 10^6 \times \frac{Z}{n} \text{ m/s}$$

Also,

$$KE = \frac{1}{2} m v^2$$

$$\Rightarrow KE = \frac{1}{2} m \times \frac{z^2 e^4}{4n^2 h^2 \epsilon_0^2}$$

$$\Rightarrow KE = \frac{1}{2} \frac{m z^2 e^4}{4n^2 h^2 \epsilon_0^2} = - \frac{z^2 e^4 m}{8n^2 h^2 \epsilon_0^2}$$

also, $F = \frac{du}{dr}$

$u \rightarrow$ potential energy

$$\Rightarrow \int_0^u du = \int_{\infty}^r F dr$$

$$\Rightarrow u - 0 = \int_{\infty}^r \frac{1}{4\pi\epsilon_0} \frac{ze^2}{r^2} dr$$

$$\Rightarrow u = -\frac{1}{4\pi\epsilon_0} \frac{ze^2}{r} - [0]$$

$$\Rightarrow u = -\frac{ze^2}{4\pi\epsilon_0 r}$$

Also, $r = \frac{n^2 h^2 \epsilon_0}{ze^2 \pi m}$

$$\Rightarrow u = -\frac{ze^2}{4\pi\epsilon_0} \frac{ze^2 \pi m}{n^2 h^2 \epsilon_0}$$

$$\Rightarrow u = -\frac{z^2 e^4 m}{4n^2 h^2 \epsilon_0^2}$$

$$\Rightarrow TE = KE + u$$

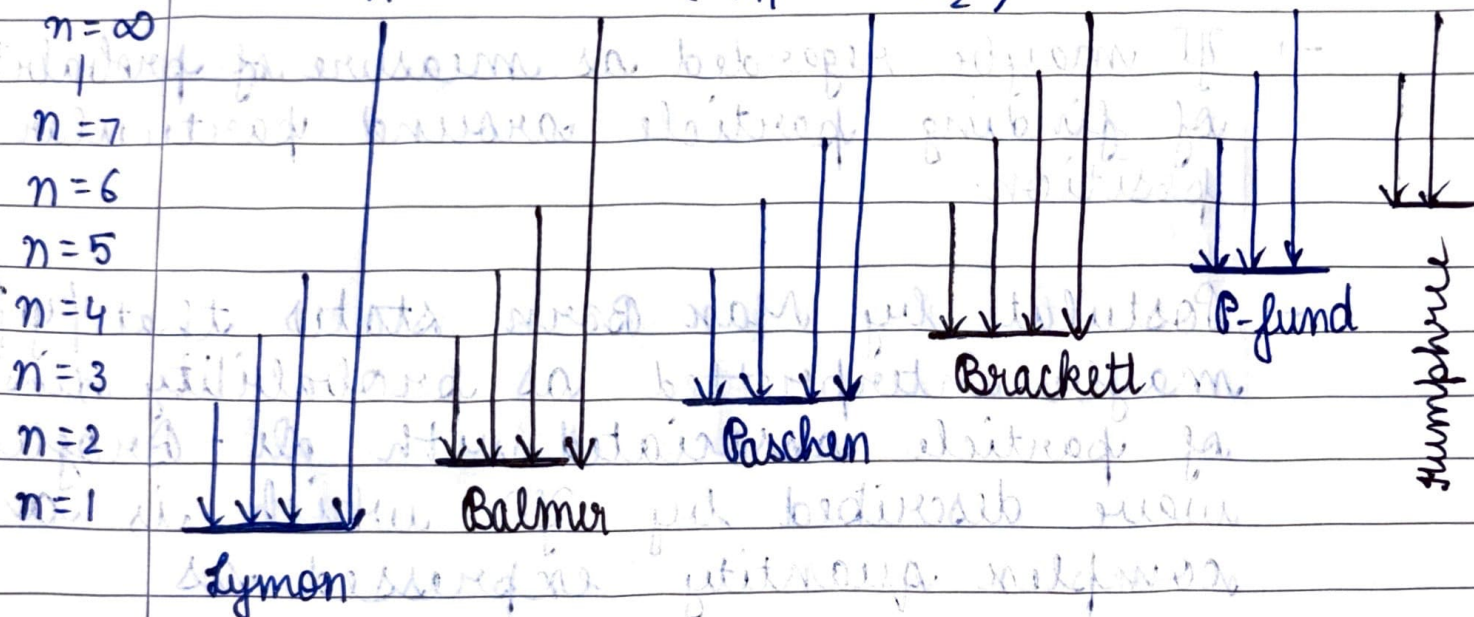
$$TE = \frac{z^2 e^4 m}{8n^2 h^2 \epsilon_0^2} - \frac{z^2 e^4 m}{4n^2 h^2 \epsilon_0^2}$$

$$\Rightarrow TE = -\frac{z^2 e^4 m}{8n^2 h^2 \epsilon_0^2}$$

$$\Rightarrow TE = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$$

Also,

$$\frac{1}{1} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$



Hydrogen spectrum
Balmer $\rightarrow$ visible Region

$$\frac{d}{dt} \psi = (t, \delta, \chi, x) \psi = (t, \delta) \psi$$

$$\psi_d + \psi_o = (d_i + o) (d_i + o) = \psi \psi = \psi$$

3. In 9 you determine hydrogen state

$$\psi_d + \psi_o = \psi (t, \delta, \chi, x) \psi = \psi$$

$$0 = \psi \quad \text{for} \quad 0 < \psi \psi =$$

→ Concept of wave packet

A wavepacket is a superposition of many waves of slightly different wavelengths and frequencies forming a localised wave.

$$\Psi(x,t) = \int A(k) e^{i(kx - \omega t)} dk$$

$A(k)$ is amplitude distribution in k -space

Wave packets allows to describe both particle and wave nature simultaneously.

→ Phase velocity (v_p)

Speed at which a single monochromatic wave phase propagates

$$v_p = \frac{\omega}{k} = \frac{\text{distance travelled by a crest}}{\text{time}}$$

→ For light in vacuum, $v_p = c$

• Characteristics

→ Can exceed speed of light in dispersive media

→ Doesn't carry energy/information

→ Associated with monochromatic waves.

$$\Psi(x,t) = A e^{i(kx - \omega t)}$$

$$k = \frac{2\pi}{\lambda}, \quad \omega = \frac{2\pi}{T}$$

Phase of wave $\phi = kx - \omega t$

at crest (constant phase)

$$\frac{d\phi}{dt} = 0$$

$$\frac{d}{dt}(Kx - \omega t) = 0$$

$$K \frac{dx}{dt} - \omega = 0$$

$$\Rightarrow \frac{dx}{dt} = v_p = \frac{\omega}{K}$$

→ Group velocity (v_g)

It's the speed at which envelope of wave packet moves which carries energy, information or particle motion.

$$v_g = \frac{d\omega}{dK}$$

→ It's actual velocity of a particle/wave packet

→ In dispersive media $v_g \neq v_p$

Consider a wave packet formed by superposition of 2 waves with diff wave nos & frequencies

$$\psi(x,t) = e^{i(K_1 x - \omega_1 t)} + e^{i(K_2 x - \omega_2 t)}$$

Since

$$e^{i\alpha} + e^{i\beta} = 2 \cos \left[\frac{\alpha - \beta}{2} \right] e^{i(\alpha + \beta)/2}$$

$$\therefore e^{i\theta} = \cos \theta + i \sin \theta$$

$$\Rightarrow \psi(x,t) = 2 \cos \left(\frac{\Delta K x - \Delta \omega t}{2} \right) e^{i(Kx - \omega t)}$$

where $K = \frac{K_1 + K_2}{2}$ $\omega = \frac{\omega_1 + \omega_2}{2}$

$\Delta K = K_2 - K_1$ $\Delta \omega = \omega_2 - \omega_1$

$\rightarrow e^{i(Kx - \omega t)} \rightarrow$ carrier wave (Phase)

$\rightarrow \cos \left[(\Delta K x - \Delta \omega t)/2 \right] \rightarrow$ envelope (group)

$\Rightarrow \frac{\Delta K x - \Delta \omega t}{2} \rightarrow$ constant

$\Rightarrow \frac{\Delta K}{2} \frac{dx}{dt} - \frac{\Delta \omega}{2} = 0$

$\Rightarrow \frac{dx}{dt} = \frac{\Delta \omega}{\Delta K}$

For $\Delta K \rightarrow 0$

$\Rightarrow V_g = \frac{d\omega}{dK}$

$\rightarrow$ Relation b/t V_g & V_p

$V_p = \frac{\omega}{K}$ $V_g = \frac{d\omega}{dK}$

For a non relativistic free particle,

$E = \frac{p^2}{2m} = \frac{(\hbar K)^2}{2m}$

Also $E = \hbar \omega$

$\Rightarrow \omega = \frac{\hbar K^2}{2m}$

$V_p = \frac{\omega}{K} = \frac{\hbar K^2}{2m K} = \frac{\hbar K}{2m}$

also,

$$v_g = \frac{d\omega}{dk} = \frac{\hbar k}{m}$$

$$\Rightarrow \underline{v_p = \frac{v_g}{2}}$$